Tuesday, April 14

- Midterm next Thursday (4/23)

If second midterm grade is higher, I will replace first midterm grade.

- No class next Tuesday (4/21)
- Office hours:
  - ↳ Canceled next Thursday (4/23)
  - ↳ Virtual Monday (4/20)
  - ↳ Extra tomorrow (4/15)

This week: Continue doing cool things with matrices, fast

Randomized Numerical Linear Algebra
- related to my research
- beautiful, heavy proof

Spectral Graph Theory Correction

$L = D - A$

Incorrect : If 0 is smallest eigenvalue of L, then 1 is top eigenvalue of A

Let $\bar{L} = D^{-1/2} L D^{-1/2}$, $\bar{A} = D^{-1/2} A D^{-1/2}$, $\bar{D} = D^{-1/2} D D^{-1/2} = I$

$\bar{L} = I - \bar{A}$ Consider $\bar{A}v = \lambda v$.

$\bar{L}v = (I - \bar{A})v = v - \lambda v = (1-\lambda)v$

Since $\bar{L}$ has eigenvalues ≥ 0, $(1-\lambda) \geq 0 \iff \lambda \leq 1$ for all λ

Since 0 is smallest eigenvalue of $\bar{L}$ with eigenvector $D^{1/2}\mathbb{1}$,

$\therefore$ 1 is top eigenvalue of $\bar{A}$ with same eigenvector

Sketched Regression

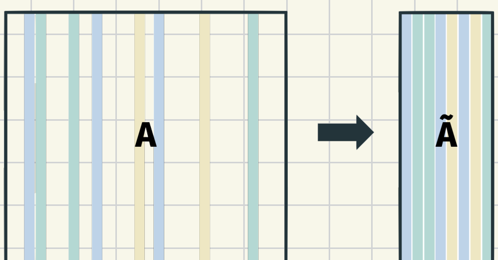
Goal: Solve least squares regression problem faster

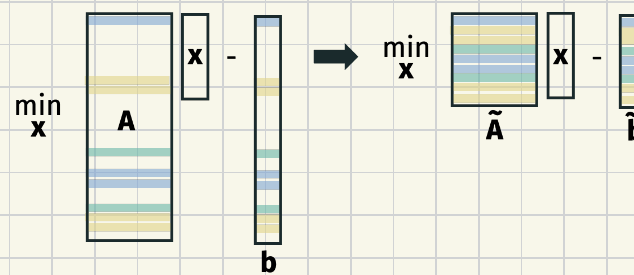
$A \in \mathbb{R}^{n \times d}$ $\quad b \in \mathbb{R}^n$

$$x^* = \arg\min_{x \in \mathbb{R}^d} \|Ax - b\|_2^2 = (A^T A)^{-1} A^T b$$

Time complexity:

Q: What if n (# cols) or d (# rows) is <u>huge</u>?





Idea: Find optimal solution to approximate problem

$$x^* = \arg\min_{x \in \mathbb{R}^d} \|Ax - b\|_2^2 \quad \text{vs.} \quad \tilde{x} = \arg\min_{x \in \mathbb{R}^d} \|\Pi A x - \Pi b\|_2^2$$

$\Pi \in \mathbb{R}^{m \times n}$ is JL projection matrix for $m \ll n$

Theorem: When $m = O(d/\epsilon^2)$ then, wp $9/10$, for any A and b,

$$\|A\tilde{x} - b\|_2^2 \leq (1+\epsilon)\|Ax^* - b\|_2^2$$

That is, $\tilde{x}$ gives almost as good of a solution as x^*

<u>Claim:</u> Under the same conditions, for all $x \in \mathbb{R}^d$,

$$(1-\epsilon)\,\|Ax-b\|_2^2 \le \|\Pi A x - \Pi b\|_2^2 \le (1+\epsilon)\,\|Ax-b\|_2^2 .$$

Proof of Theorem using claim:

Plan: Let's prove the claim :)

Distributional JL: If $m = O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$ then, for a fixed y, wp $1-\delta$,

$$(1-\epsilon)\,\|y\|_2^2 \le \|\Pi y\|_2^2 \le (1+\epsilon)\,\|y\|_2^2.$$

Challenge: JL holds for <u>one</u> y whereas claim holds for <u>infinite</u> $Ax-b$

Idea: Yes, $Ax-b \in \mathbb{R}^n$ <u>but</u> lies in d-dimensional subspace (i.e., linear combo of $d+1$ vecs).

Subspace Embedding: Let $\mathcal{U} \subset \mathbb{R}^n$ be a d-dim subspace.

If $m = O\left(\frac{d\log(1/\epsilon) + \log(1/\delta)}{\epsilon^2}\right)$ then, wp $1-\delta$, for all $y \in \mathcal{U}$,

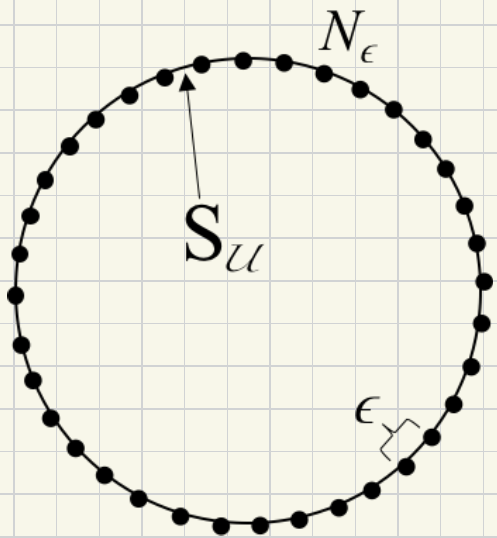
$$(1-\epsilon)\|y\|_2^2 \leq \|\Pi y\|_2^2 \leq (1+\epsilon)\|y\|_2^2.$$

Notice the claim follows since $\mathcal{U} = \{Ax - b : \forall x \in \mathbb{R}^d\}$ is $(d+1)$-dim.

Notice it suffices to prove the embedding result for w with $\|w\|_2^2 = 1$ since the statement is scale-invariant.

Let $S_{\mathcal{U}} = \{w \in \mathcal{U} : \|w\|_2 = 1\}$

Epsilon-Net : There are not too many nearby points on unit sphere

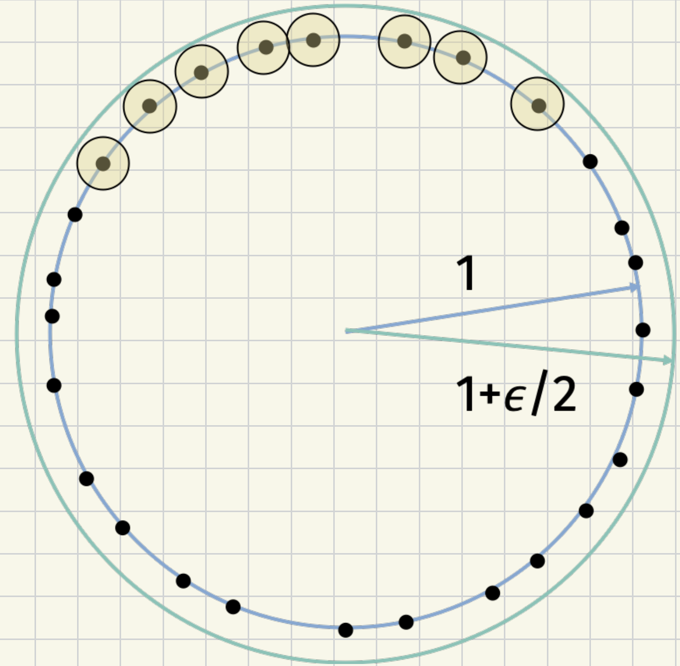


For $\epsilon < 1$, there exists $\mathcal{N}_\epsilon \subset S_u$ with

① $|\mathcal{N}_\epsilon| \leq \left(\frac{3}{\epsilon}\right)^d$

② For all $v \in S_u$, $\min_{w \in \mathcal{N}_\epsilon} \|v - w\|_2 \leq \epsilon$

(Hypothetical) Epsilon-Net Construction:



Greedily add points that are at least ϵ-away from all previously added points.

② clearly holds, let's prove ① ...

Recall $\text{vol}(d, r) = c_d r^d$.

$$\text{vol}(d, \tfrac{\epsilon}{2})\, |\mathcal{N}_\epsilon| \leq \text{vol}(d, 1 + \tfrac{\epsilon}{2})$$

$$\Rightarrow |\mathcal{N}_\epsilon| \leq \frac{(1+\frac{\epsilon}{2})^d}{(\epsilon/2)^2} = \left(\frac{2+\epsilon}{2} \cdot \frac{2}{\epsilon} \right)^d \leq \left(\frac{3}{\epsilon}\right)^d$$